

R Advanced Data Analysis Course - syllabus

Ronnie Bailey-Steinitz

2026-02-26

6-Week Advanced Data Analysis in R: Choosing & Applying Statistical Models

Pre-Course Refresher Workshop (led by Ronnie)

Goal: Refresh core wrangling, visualization, and modeling skills before advanced analysis.

Week 1: Basics (May 29, 2026)

- Set up a new project and find working directory
- Initiate a new Rmd. file
- Import CSV with `readr::read_csv()`
- Exploratory stats and plots `summary()`, `glimpse()`, `hist()`
- Review visualization: `ggplot()`, `geom_point()`, `geom_boxplot()`, `geom_histogram()`

Week 2: Wrangling (June 5, 2026)

- Review wrangling: `dplyr::select()`, `dplyr::filter()`, `dplyr::mutate()`, `dplyr::group_by()`, `dplyr::summarise()`
- Joining datasets with `left_join()`
- Linear models review: `lm()`, `summary()`
- Transformations: `log()`, `scale()`
- **Lab Challenge** Clean, visualize, and briefly interpret one variable from your own dataset

Week 1: From Research Question to Statistical Model

June 12, 2026

Dataset: Hormone + behavioral dataset

- Identify variable types: continuous, count, binary, categorical, proportion
- Independence vs. repeated measures
- Cross-sectional vs. longitudinal structure
- Diagnose distributions with `geom_histogram()`, `geom_density()`, `qqnorm()`, `qqline()`
- Inspect structure using `str()`, `summary()`

- Build a statistical decision framework
 - **Visualization:** distribution plots and grouped summaries
 - **Lab Challenge:** Identify the correct model type for 3 different biological questions from your dataset
-

Week 2: Comparing Groups – t-tests, ANOVA & Nonparametric Alternatives

June 19, 2026

Dataset: Hormone dataset

- Two-sample `t.test()`
 - Paired `t.test()`
 - ANOVA with `aov()`
 - Check assumptions: normality, homogeneity of variance
 - Nonparametric alternatives: `wilcox.test()`, `kruskal.test()`
 - Effect sizes and biological interpretation
 - **Visualization:** `geom_boxplot()`, `geom_violin()`, `geom_jitter()`
 - **Lab Challenge:** Compare hormone levels across demographic or health categories
-

Week 3: Count Data, Skewed Data & Zero Inflation

June 26, 2026

Dataset: Aggression counts, parasite load, vert census data

- When normality is inappropriate
- Poisson regression with `glm(family = "poisson")`
- Detecting overdispersion
- Negative binomial models with `MASS::glm.nb()`
- Conceptual introduction to zero-inflated models

- Gamma models for right-skewed continuous data
 - When (and when not) to apply `log()` transformations
 - **Visualization:** predicted values with `geom_line()` and `geom_point()`
 - **Lab Challenge:** Fit Poisson and negative binomial models and compare model fit
-

Week 4: Binary & Categorical Outcomes – Chi-Square & Logistic Regression

July 10, 2026

Dataset: Health status, survival, demographic transitions

- Contingency tables with `table()`
 - `chisq.test()`
 - `fisher.test()`
 - Logistic regression with `glm(family = "binomial")`
 - Interpreting odds ratios
 - Generate predicted probabilities
 - **Visualization:** probability curves using `geom_line()`
 - **Lab Challenge:** Model probability of sickness using age, sex, or rank
-

Week 5: Mixed Effects Models – Repeated Measures & Hierarchical Data

July 17, 2026

Dataset: Longitudinal dataset (environmental measurement?)

- Why repeated measures violate independence
- Random intercept models with `lme4::lmer()`
- Conceptual introduction to random slopes
- Individual ID as random effect

- Nested data structure (individual within group)
 - Compare mixed vs. non-mixed models
 - Inspect model output with `summary()`
 - Extract fitted values for visualization
 - **Visualization:** predicted effects with `geom_line()`
 - **Lab Challenge:** Fit a mixed model with repeated hormone data and interpret fixed vs. random effects
-

Week 6: Model Evaluation, Interpretation & Scientific Communication

July 24, 2026

Dataset: Student-selected dataset

- Residual diagnostics
 - Model comparison using AIC
 - Interpreting interaction terms
 - Confidence intervals vs. p-values
 - Visualizing model predictions
 - Reporting analyses clearly in RMarkdown
 - **Final Project Lab:**
 - Define biological question
 - Justify model choice
 - Run analysis
 - Visualize results
 - Interpret findings
-

Final Presentation Session

July 31, 2026

- Each student gives 5–7 minute presentation
- Emphasis on:
 - Clear biological question
 - Appropriate model choice
 - Assumptions considered
 - Interpretation beyond p-values
 - Clean, publication-quality visualization